



School of Technology Course Catalog

Prepare for the future through hands-on technology courses that build real world skills in programming, cybersecurity, information technology, web development, and artificial intelligence. Whether you're exploring technology for the first time or preparing for an industry certification exam, there's a pathway designed for you.

Every course is **one semester long** and **asynchronous**, allowing students to work at a pace that fits their schedule. Students choose when during the week they work and may work ahead, but each course has weekly pacing targets and must be completed by the end of the semester. Plan on about 45 minutes of coursework every school day.

Two **recommended 30-minute video sessions** are held every Friday, one for grades 6-8 and one for grades 9-12, open to students in any course. They are not required — they are time to share projects, connect with other students, and build community around emerging technology. For help with coursework, students should visit instructor office hours on Monday, Tuesday, and Thursday from 9:00 to 10:30 a.m.

Each course listing includes its semester availability.

S1 – offered only during Fall semester.

S2 – offered only during Winter semester.

S1 or S2 – offered during either semester.



QUESTIONS? GET IN TOUCH

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See the **School of Technology Program Guide** for pacing, support, and expectations.



ABOUT CERTIFICATION EXAMS

Certification exams are administered by CompTIA and TestOut, scheduled separately from the course, and paid for by the student or family. The

instructor can help students prepare and register.

Foundations of Programming



Formerly Exploring Tech I

Exploring Technology and Art

S1

GRADES 6-12

Combine creativity with computer science to build your own interactive games and music. You'll begin using Code.org's Game Lab environment to program shapes, sprites, and animations, handle user input, and use variables and conditionals to turn simple images into complex, multi-level games.

Then you'll explore foundational programming concepts through music, using code to build, remix, and perform original songs. Through sequencing, loops, and functions, you'll practice computational thinking like pattern recognition, algorithmic thinking, and debugging to design, test, and refine your musical ideas.



Formerly Exploring Tech II

Intro to Python

S2

GRADES 6-12

An introduction to the fundamentals of computer science through visual programming and text-based logic. In CodeHS you'll master top-down design, functions, and decomposition to solve grid-based puzzles and build a strong foundation in algorithmic thinking and problem-solving.

As you transition to standard Python, you'll manage data with variables, user input, and mathematical operators, then master control structures including if/else statements and logical operators. The course concludes with looping and nested logic, where you'll build interactive projects like a Mad Libs generator, a Quiz Game, and a Password Authenticator.



Intro to Cybersecurity

S1 OR S2**GRADES 9-12**

Start with the digital world you already live in. You'll look at what your online life reveals about you, how companies collect and use your personal data, how to judge whether a source can be trusted, and where ethics and law draw the line online, then learn to recognize phishing, malware, and other common attacks and how to stop them.

From there you'll make and break codes, from the Caesar wheel to public-key encryption and hashing, and build a cipher-driven escape room of your own. The semester finishes with the machine and the network: operating systems, user accounts and permissions, the command line, and how a message travels the internet through binary, packets, DNS, and the OSI model. No prior computer science background is needed.



Ethical Hacker

S1 OR S2**GRADES 9-12**

Learn the professional process used to test and secure modern digital networks. You'll start where a real engagement starts: the ethics and law of authorized testing, then scoping a job and writing the rules of engagement. From there you'll gather open-source intelligence on a target, scan and enumerate it with Nmap, and assess what you find using the CVE database and CVSS scoring.

Then you'll exploit a known vulnerability on a sandboxed lab machine, escalate privileges, harvest credentials, and move laterally, and study how attackers persist and cover their tracks, and how a defender detects them. The course closes with a written exam and a full attack-lifecycle lab practical.



CyberDefense Pro

S1 OR S2

GRADES 9-12

PREREQUISITE

Pass Intro to Cybersecurity or Ethical Hacker first.

An advanced course that prepares you for a career as a cybersecurity analyst. You'll detect and respond to modern threats by mastering vulnerability management, threat hunting, and network security, using professional tools like SIEMs and protocol analyzers to monitor traffic, identify indicators of compromise, and defend against attacks like SQL injection and ransomware.

With an emphasis on incident response and digital forensics, you'll learn to contain breaches and analyze evidence, and explore specialized security for cloud, wireless, and IoT systems, all while preparing for the CompTIA CySA+ (CS0-003) and TestOut CyberDefense Pro certification exams.

TRACK

IT & Hardware

This track is a strict three-course sequence: IT Fundamentals, then PC Pro A, then PC Pro B.



IT Fundamentals

S1 OR S2

GRADES 9-12

A broad introduction to the modern world of technology, preparing you for the CompTIA Tech+ certification exam. You'll start with computing basics, data privacy, and internet technologies, then cover the essential networking and cybersecurity concepts that keep data moving safely across the globe.

From there you'll dive into hardware, software, and databases to understand how computers are built and store information, get a head start in coding and logic, and look ahead at emerging technologies including AI, robotics, and virtual reality.



PC Pro A

S1

GRADES 9-12

PREREQUISITE

Pass IT Fundamentals first.

A comprehensive introduction to IT that prepares you for industry certification exams like TestOut PC Pro and CompTIA A+. You'll learn the core responsibilities of a PC technician, using professional ticketing systems, managing documentation, and following safety and maintenance procedures.

The focus is hands-on hardware: installing, configuring, and troubleshooting motherboards, processors, and power supplies. You'll explore Windows, Linux, and macOS, manage storage systems like RAID and partitioning, and master system implementation by performing full Windows installations and building virtual machines.



PC Pro B

S2

GRADES 9-12

PREREQUISITE

Pass PC Pro A first.

The advanced skills needed to manage and protect a professional IT environment. You'll master Windows and Linux management through system tools, scripting, and the command line, including managing users in Active Directory, setting up VPNs for remote work, and performing critical data backups.

You'll also explore networking and mobile support, configuring secure Wi-Fi and troubleshooting devices from laptops to 3D printers, then finish with a focus on cybersecurity: configuring firewalls, blocking malware, and defending against social engineering to keep an organization's data safe.

Web Design and Development



Web Design and Dev I

S1

GRADES 9-12

Learn the foundational skills to create, style, and publish your own professional websites. You'll explore how web pages are developed and delivered across the globe, then start building right away with HTML, mastering tags, attributes, and elements to structure pages with images, hyperlinks, and data tables.

Next you'll style your creations with CSS, using selectors and classes to manage layout, color, and fonts and understanding how the “cascade” sets visual priority. You'll design a live homepage portfolio from scratch, then finish with advanced techniques: divs and spans for precise layout, embedded media through iframes, and CSS animations that make sites dynamic.



Web Design and Dev II

S2

GRADES 9-12

PREREQUISITE

Pass Web Design and Dev I first.

Transition from basic page construction to building professional, responsive, user-centered experiences. You'll start with the Bootstrap framework, using open-source components and grid systems to create mobile-responsive layouts that adapt to any screen, while practicing how to read and implement real technical documentation.

The focus then shifts to UI and UX design: researching stakeholder needs, defining clear problem statements, rapid prototyping from paper sketches to digital solutions, and user testing with peers. The course concludes with a comprehensive final project where you design, prototype, and develop a website from scratch.

Emerging Technology



Artificial Intelligence

S1 OR S2**GRADES 9-12**

AI Foundations introduces you to the core principles of computer science and the growing role of artificial intelligence in everyday life. Through hands-on programming, visual problem solving, and critical analysis, you'll explore how computing systems work, how data powers AI, and how intelligent tools make decisions.

The semester moves through six units: Problem Solving with AI, Foundations of AI Programming, AI and the Systems That Power It, The Fabric of the Internet and AI, AI-Powered Threats and Defenses, and Insights from Data and AI. Each one builds real skills, from writing Python to analyzing cybersecurity risks, and ends with a project where you apply what you've learned.

Questions? muenchen@miprepschool.org